

Activity 2.4.1

ST10440289

```
4 namespace WildlifeApp
5 {
6     class Program
7     {
8         static void Main(string[] args)
9         {
10             List<Animal> animals = new List<Animal>();
11             bool running = true;
12
13             while (running)
14             {
15                 Console.WriteLine("\nWildlife App Menu:");
16                 Console.WriteLine("1. Add Bird");
17                 Console.WriteLine("2. Add Butterfly");
18                 Console.WriteLine("3. View All Sightings");
19                 Console.WriteLine("4. Exit");
20                 Console.WriteLine("Choose an option (1-4): ");
21                 string choice = Console.ReadLine();
22
23                 switch (choice)
24                 {
25                     case "1":
26                         Bird bird = new Bird();
27                         Console.WriteLine("Enter the English name of the bird: ");
28                         bird.EnglishName = Console.ReadLine();
29                         Console.WriteLine("Enter the Latin name of the bird: ");
30                         bird.LatinName = Console.ReadLine();
31                         bird.DateLastSighted = GetValidDate("Enter the date last sighted (yyyy-mm-dd): ");
32                         animals.Add(bird);
33                         break;
34
35                     case "2":
36                         Butterfly butterfly = new Butterfly();
37                         Console.WriteLine("Enter the English name of the butterfly: ");
38                         butterfly.EnglishName = Console.ReadLine();
39                         Console.WriteLine("Enter the Latin name of the butterfly: ");
40                         butterfly.LatinName = Console.ReadLine();
41                         butterfly.DateLastSighted = GetValidDate("Enter the date last sighted (yyyy-mm-dd): ");
42                         animals.Add(butterfly);
43                         break;
44
45                     case "3":
46                         Console.WriteLine("\n--- All Sightings ---");
47                         foreach (Animal animal in animals)
48                         {
49                             animal.DisplaySighting();
50                         }
51                         break;
52
53                     case "4":
54                         running = false;
55                         break;
56
57                     default:
58                         Console.WriteLine("Invalid option. Please try again.");
59                         break;
60                 }
61             }
62         }
63
64         static DateTime GetValidDate(string prompt)
65         {
66             DateTime date;
67             Console.WriteLine(prompt);
68             while (!DateTime.TryParse(Console.ReadLine(), out date))
69             {
70                 Console.WriteLine("Invalid date format. Try again (yyyy-mm-dd): ");
71             }
72             return date;
73         }
74     }
75 }
76
```

```
47 foreach (Animal animal in animals)
48 {
49     animal.DisplaySighting();
50 }
51 break;
52
53 case "4":
54     running = false;
55     break;
56
57 default:
58     Console.WriteLine("Invalid option. Please try again.");
59     break;
60 }
61 }
62
63
64 2 references
65 static DateTime GetValidDate(string prompt)
66 {
67     DateTime date;
68     Console.WriteLine(prompt);
69     while (!DateTime.TryParse(Console.ReadLine(), out date))
70     {
71         Console.WriteLine("Invalid date format. Try again (yyyy-mm-dd): ");
72     }
73     return date;
74 }
75
76
```

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6
7 namespace WildlifeApp
8 {
9     interface ISighting
10     {
11         void DisplaySighting();
12     }
13 }
14
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace WildlifeApp
{
    class Butterfly : Animal
    {
        public override void displaySighting()
        {
            Console.WriteLine($"[Butterfly] English Name: {EnglishName}, Latin Name: {LatinName}, Last Sighted: {DateLastSighted.ToShortDateString()}");
        }
    }
}
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace WildlifeApp
{
    class Bird : Animal
    {
        public override void displaySighting()
        {
            Console.WriteLine($"[Bird] English Name: {EnglishName}, Latin Name: {LatinName}, Last Sighted: {DateLastSighted.ToShortDateString()}");
        }
    }
}
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace WildlifeApp
{
    abstract class Animal : ISighting
    {
        public string EnglishName { get; set; }
        public string LatinName { get; set; }
        public DateTime DateLastSighted { get; set; }
        public abstract void displaySighting();
    }
}
```